Dynamical Friction in Fuzzy Dark Matter Halos: Quantum Pressure Constraints on Stellar Orbits

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Abstract

This paper investigates the suppression of dynamical friction in Fuzzy Dark Matter (FDM) halos and its consequences for the orbital stability of massive perturbers such as globular clusters. Modeling the dark matter halo as a coherent scalar field governed by the Schrödinger-Poisson equations, we recast the wave dynamics in Madelung hydrodynamical form and identify the quantum pressure term responsible both for the smooth solitonic core structure of FDM halos and for its departure from classical Chandrasekhar dynamical friction. Using linear perturbation theory and the Born approximation, we derive the density wake generated by a moving mass and show that the resulting drag force reduces to the classical $1/v^2$ Chandrasekhar result at high velocity but is suppressed to scale linearly with v once the perturber's velocity falls below the local de Broglie velocity.

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Contents

1 Introduction 1
1.1 The Cold Dark Matter Paradigm and its Small-Scale Crises 1
1.2 The Ultralight Axion (Fuzzy Dark Matter) Hypothesis 2
2 Wave-Mechanical Formulation of FDM 3
2.1 The Schrödinger-Poisson System 3
2.2 Madelung Hydrodynamics 4
2.3 The Quantum Pressure Tensor 4
3 Linear Perturbation Theory of the Wave Wake 6
3.1 Unperturbed Background 6
3.2 Linearized Wave Equations 6
3.3 The Density Wake and Interference Fringes 7
4 Quantum-Suppressed Dynamical Friction 7
4.1 Force Integration 7
4.2 The High-Velocity Limit ($v \gg v_{\mathrm{dB}}$) 8
4.3 The Low-Velocity Limit ($v \lesssim v_{\mathrm{dB}}$) 9
5 Orbital Dynamics in Solitonic Cores 10
5.1 Hamiltonian Formulation of the Orbit 10
5.2 Numerical Integration of Orbital Decay 11
5.3 Derivation of the Stalling Radius (r_{stall}) 12
6 Observational Constraints: The Fornax Dwarf Spheroidal . 13
6.1 Globular Cluster Kinematics in Fornax 13
6.2 Applying the FDM Stalling Model 14
6.3 Constraining the Boson Mass 15
7 Conclusion 15
7.1 Summary of Analytical Findings 15
7.2 Cosmological Implications 16

1. Introduction

1.1 The Cold Dark Matter Paradigm and its Small-Scale Crises

The standard cosmological model, ΛCDM, has achieved remarkable success in reproducing the large-scale structure of the Universe (> 1 Mpc). From the power spectrum of cosmic microwave background anisotropies to the filamentary distribution of galaxies on Megaparsec scales, cold dark matter (CDM) modeled as a collisionless, pressureless fluid of heavy particles ($m \sim$ GeV–TeV) provides a robust dynamical framework.

However, on sub-galactic scales ($\lesssim 10$ kpc), pure N-body collisionless dark matter simulations systematically diverge from astronomical observations. These small-scale discrepancies broadly encompass three classical problems:

1. **The Cusp-Core Problem:** Cosmological N-body simulations (e.g., Navarro-Frenk-White or NFW profiles) predict that dark matter halos should exhibit steep central density profiles ($\rho(r) \propto r^{-1}$). In contrast, high-resolution kinematic observations of low-surface-brightness galaxies and dwarf spheroidals (dSphs) reveal extended, constant-density central cores ($\rho(r) \propto r^{0}$).

2. **The Missing Satellites and "Too-Big-To-Fail" Problems:** Collisionless CDM models overpredict the abundance of sub-halos around Milky Way-sized galaxies by roughly an order of magnitude compared to observed dwarf satellites, while predicting central densities for those sub-halos that are significantly higher than observed.

3. **The Globular Cluster Timing Problem:** In the Fornax dwarf spheroidal galaxy, five globular clusters orbit within 1.6 kpc of the galactic center. According to classical Chandrasekhar dynamical friction theory, a massive object moving through a background of collisionless DM particles continuously loses orbital angular momentum to its gravitational wake. For realistic mass profiles, the orbital decay timescale for the Fornax clusters is:

$$t_{\rm df} \sim 2\text{–}3 \text{ Gyr}$$

Because Fornax is an ancient system (> 10 Gyr), these clusters should have long since spiraled into the galactic center and coalesced into a single nucleus. The observed survival of these clusters at finite

non-zero radii indicates a fundamental breakdown in standard Chandrasekhar dynamical friction within low-mass galactic halos.

While baryonic feedback mechanisms (such as supernova-driven gas outflows) have been proposed to flatten central dark matter cusps and alter cluster orbits, their efficiency in low-luminosity, star-poor dwarf spheroidals remains highly contested. This has motivated alternative dark matter candidates whose intrinsic physical properties naturally resolve these small-scale crises.

1.2 The Ultralight Axion (Fuzzy Dark Matter) Hypothesis

Fuzzy Dark Matter (FDM)—also termed Ultralight Dark Matter (ULDM), replaces heavy, collisionless WIMP particles with an extremely light scalar boson (such as an axion-like particle) with a mass on the order of:

$$m \sim 10^{-22}\ \mathrm{eV}/c^2$$

At this mass scale, the macroscopic de Broglie wavelength of the dark matter particle reaches astronomical scales:

$$\lambda_{\mathrm{dB}} = \frac{h}{mv} \approx 1.2\ \mathrm{kpc} \left(\frac{10^{-22}\ \mathrm{eV}}{m}\right) \left(\frac{100\ \mathrm{km/s}}{v}\right)$$

Because the phase-space occupation number within a cubic de Broglie wavelength is extraordinarily large ($N \gg 1$), the individual particle identities overlap, and the system behaves non-relativistically as a single coherent quantum scalar field $\psi(\mathbf{r}, t)$—a Bose-Einstein condensate governed by the Schrödinger-Poisson equations.

- **On cosmological scales ($\mathbf{r \gg \lambda_{dB}}$):** The wave properties average out, and FDM is functionally indistinguishable from standard ΛCDM, retaining all large-scale cosmological successes.
- **On galactic scales ($\mathbf{r \lesssim \lambda_{dB}}$):** Quantum mechanical effects dominate. A gradient in the field amplitude generates an effective "quantum pressure" that opposes gravitational collapse below a characteristic quantum Jeans length λ_{J}.

This quantum pressure produces two crucial features:

1. It replaces the singular NFW cusp with a smooth, thermodynamically stable, constant-density central soliton core.

2. It fundamentally alters the gravitational wake generated by moving bodies. When a massive perturber moves at or below the local de Broglie phase velocity ($v \lesssim v_{\rm dB} \equiv \hbar/m\lambda_{\rm dB}$), quantum pressure suppresses small-scale density perturbations in the background field, causing dynamical friction to stall.

2. Wave-Mechanical Formulation of FDM

2.1 The Schrödinger-Poisson System

In the standard formulation of Fuzzy Dark Matter (FDM), the dark matter sector consists of a real scalar field ϕ minimally coupled to gravity, governed relativistically by the Einstein-Klein-Gordon equations. Because the virial velocities in galactic halos are highly non-relativistic ($v \sim 10^{-3}c$) and the scalar particle mass is ultralight ($m \sim 10^{-22}$ eV), the scalar field dynamics reduce to their Newtonian limit.

We extract the rapidly oscillating rest-mass phase from the real scalar field to define a complex, slowly varying macroscopic wave function $\psi(\mathbf{r}, t)$:

$$\phi(\mathbf{r}, t) = \frac{1}{\sqrt{2m}} \left(\psi(\mathbf{r}, t) e^{-imt/\hbar} + \psi^*(\mathbf{r}, t) e^{imt/\hbar} \right)$$

Under the weak-field, non-relativistic approximation, the dynamics of ψ are governed by the coupled Schrödinger-Poisson system. The field evolves according to the free Schrödinger equation embedded in a self-consistent gravitational potential $\Phi(\mathbf{r}, t)$:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + m\Phi\psi$$

The wave function is normalized such that the macroscopic dark matter mass density is given by $\rho(\mathbf{r}, t) = m|\psi(\mathbf{r}, t)|^2$. The gravitational potential is thus sourced by the density via the classical Poisson equation:

$$\nabla^2 \Phi = 4\pi G\rho = 4\pi G m |\psi|^2$$

Unlike a single-particle quantum state, ψ describes a Bose-Einstein condensate encompassing an astronomical number of bosons. The deterministic, classical nature of this field on macroscopic scales ensures that ρ acts as the continuous fluid density of the dark matter halo.

2.2 Madelung Hydrodynamics

To analyze dynamical friction and structural formation, it is advantageous to translate the wave mechanics of the Schrödinger-Poisson system into a hydrodynamical framework. We apply the Madelung transformation, decomposing the complex wave function into a real amplitude and a real phase $S(\mathbf{r}, t)$:

$$\psi(\mathbf{r}, t) = \sqrt{\frac{\rho(\mathbf{r}, t)}{m}} \exp\left(i \frac{S(\mathbf{r}, t)}{\hbar}\right)$$

We define the macroscopic fluid velocity field u(r,t) as the gradient of the phase:

$$\mathbf{u} = \frac{\nabla S}{m}$$

Because the velocity field is a pure gradient ($\nabla \times \mathbf{u} = 0$), the FDM fluid is strictly irrotational (except at topological nodes where $\rho = 0$, leading to quantized vortices).

Substituting the polar form of ψ into the Schrödinger equation and separating the real and imaginary components yields two equations of state. The imaginary part directly generates the standard classical continuity equation, guaranteeing the conservation of dark matter mass:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

This confirms that despite its quantum mechanical origins, FDM adheres to the fundamental kinematic mass conservation laws expected of any dark matter fluid.

2.3 The Quantum Pressure Tensor

The real part of the decomposed Schrödinger equation defines the phase evolution, yielding a generalized Hamilton-Jacobi equation. Taking the gradient of this equation and dividing by the particle mass m produces the quantum-modified Euler equation:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla \Phi - \frac{1}{m} \nabla Q$$

The left side of the equation is the standard convective derivative of velocity, while the right side introduces the forces acting on the fluid. The first term, $-\nabla \Phi$, represents classical gravitational free-fall.

The critical departure from Cold Dark Matter (ΛCDM) is the second term, which is sourced by the Quantum Pressure Potential Q:

$$Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}$$

This term prevents the singular collapse of dark matter. To understand its mechanical effect, we can recast the gradient of Q in the form of a classical Cauchy stress tensor. The effective quantum pressure tensor σ_{ij} is defined such that $\frac{1}{m}\rho\partial_i Q = \partial_j \sigma_{ij}$, taking the form:

$$\sigma_{ij} = \frac{\hbar^2}{4m^2}\rho\frac{\partial^2 \ln\rho}{\partial x_i \partial x_j}$$

Unlike classical thermal pressure, which is isotropic and depends on local temperature, the FDM quantum pressure is highly anisotropic and scales with the local curvature of the density field. It specifically opposes sharp, localized density gradients.

When gravity attempts to collapse FDM into a sharp density cusp ($\nabla^2\sqrt{\rho} \gg 0$), Q generates a violent repulsive gradient force. This balancing act between Newtonian gravity and quantum pressure establishes a fundamental minimum length scale for gravitational collapse. Linearizing the quantum Euler-Poisson system about a uniform background density ρ gives the dispersion relation for a plane-wave perturbation of wavenumber k:

$$\omega^2 = \frac{\hbar^2 k^4}{4m^2} - 4\pi G\rho$$

Setting $\omega^2 = 0$ defines the critical Jeans wavenumber $k_J = \left(16\pi G\rho m^2/\hbar^2\right)^{1/4}$, and hence the Quantum Jeans length $\lambda_J \equiv 2\pi/k_J$:

$$\lambda_J \approx \left(\frac{\pi^3\hbar^2}{G\rho m^2}\right)^{1/4}$$

Perturbations on scales smaller than λ_J are stabilized by quantum pressure and dissipate as sound waves, whereas scales larger than λ_J undergo standard Jeans collapse. In galactic centers, this equilibrium manifests as the solitonic core—a stable, flat-density region that completely replaces the sharp NFW cusp, effectively resolving the first of the small-scale cosmological crises.

3. LINEAR PERTURBATION THEORY OF THE WAVE WAKE

3.1 Unperturbed Background

To isolate the effects of dynamical friction, we transition to the rest frame of the massive perturber M (e.g., a globular cluster). In this frame, the perturber is fixed at the origin, and the background Fuzzy Dark Matter (FDM) halo behaves as a continuous fluid flowing past the mass with an asymptotic uniform velocity $\mathbf{v} = v\hat{\mathbf{z}}$.

Assuming the background density ρ_0 is locally constant on scales smaller than the core radius, the unperturbed FDM state is described by a macroscopic plane wave:

$$\psi_0(\mathbf{r}) = \sqrt{\frac{\rho_0}{m}} e^{i\mathbf{k}\cdot\mathbf{r}}$$

where the wave vector is defined by the de Broglie relation $\mathbf{k} = m\mathbf{v}/\hbar$. The kinetic energy of this background flow is $E = \frac{\hbar^2 k^2}{2m} = \frac{1}{2}mv^2$.

3.2 Linearized Wave Equations

The presence of the perturber introduces a localized external gravitational potential $\Phi_{\text{ext}}(\mathbf{r}) = -GM/r$. This potential scatters the incoming FDM wave, generating a perturbed wave function:

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) + \psi_1(\mathbf{r})$$

where ψ_1 represents the scattered wave component.

Because we are modeling the steady-state gravitational wake, we utilize the time-independent Schrödinger equation:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + m\Phi_{\text{ext}}(\mathbf{r}) + m\Phi_{\text{self}}(\mathbf{r})\right)\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

To strictly isolate the kinematic drag generated by the perturber, we assume the perturbation is weak ($|\psi_1| \ll |\psi_0|$) and neglect the self-gravity of the FDM wake ($\Phi_{\text{self}} \approx 0$), an approximation valid for high-velocity or low-mass perturbers. Substituting the perturbed wave function into the Schrödinger equation and retaining only first-order terms yields a driven Helmholtz equation:

$$(\nabla^2 + k^2)\psi_1(\mathbf{r}) = \frac{2m^2}{\hbar^2}\Phi_{\text{ext}}(\mathbf{r})\psi_0(\mathbf{r})$$

The right side acts as a source term, where the classical gravitational potential of the moving mass distorts the incoming quantum phase.

3.3 The Density Wake and Interference Fringes

We solve this inhomogeneous differential equation using the standard free-space Green's function for the Helmholtz operator, $G(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi}\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}$. Applying the first-order Born approximation, the scattered wave function is:

$$\psi_1(\mathbf{r}) = -\frac{2m^2}{\hbar^2}\int \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}\left(-\frac{GM}{|\mathbf{r}'|}\right)\psi_0(\mathbf{r}')d^3\mathbf{r}'$$

The physical observable of interest is the macroscopic dark matter density, $\rho(\mathbf{r}) = m|\psi|^2$. Expanding this to first order in the perturbation, the density wake $\delta\rho(\mathbf{r}) = \rho(\mathbf{r}) - \rho_0$ is given by the interference between the incoming plane wave and the scattered wave:

$$\delta\rho(\mathbf{r}) = m\left(\psi_0^*\psi_1 + \psi_1^*\psi_0\right) = 2m\,\mathrm{Re}(\psi_0^*\psi_1)$$

Evaluating the integral yields the explicit spatial structure of the FDM gravitational wake:

$$\frac{\delta\rho(\mathbf{r})}{\rho_0} = \frac{2GM}{v^2 r}\sin\left(kr(1-\cos\theta)\right)$$

where $r = |\mathbf{r}|$ is the distance from the perturber and θ is the scattering angle relative to the velocity axis $\hat{\mathbf{z}}$.

4. QUANTUM-SUPPRESSED DYNAMICAL FRICTION

4.1 Force Integration

The dynamical friction force experienced by the mass M is the net gravitational attraction exerted back on it by the distorted background density field $\delta\rho(\mathbf{r})$. By Newton's law of universal gravitation, the net force vector is given by the spatial integral over the entire halo volume:

$$\mathbf{F}_{\mathrm{drag}} = \int_V GM\frac{\delta\rho(\mathbf{r})}{r^2}\hat{\mathbf{r}}\,d^3\mathbf{r}$$

By virtue of the azimuthal symmetry of the density wake about the stream velocity axis $\hat{\mathbf{z}}$, all transverse force components cancel ($\int \delta\rho \sin\theta\cos\phi\, d^3\mathbf{r} =$

0). The net force acts strictly antiparallel to the perturber's velocity vector $\mathbf{v} = v\hat{\mathbf{z}}$:

$$\mathbf{F}_{\rm drag} = -F_{\rm drag}\hat{\mathbf{z}}, \quad F_{\rm drag} = -GM \int_{r_{\rm min}}^{r_{\rm max}} \int_0^{\pi} \int_0^{2\pi} \delta\rho(r,\theta) \cos\theta \sin\theta \, dr \, d\theta \, d\phi$$

Substituting the wave interference density wake derived in Section 3, $\delta\rho(r,\theta) = \rho_0 \frac{2GM}{v^2 r} \sin\left(kr(1-\cos\theta)\right)$, the integrand becomes:

$$F_{\rm drag} = -\frac{4\pi G^2 M^2 \rho_0}{v^2} \int_{r_{\rm min}}^{r_{\rm max}} \frac{dr}{r} \int_0^{\pi} \cos\theta \sin\theta \sin\left(kr(1-\cos\theta)\right) d\theta$$

To evaluate the angular integral, we introduce the substitution $x = 1-\cos\theta$, so $dx = \sin\theta d\theta$ and $\cos\theta = 1 - x$. As θ ranges from 0 to π, x ranges from 0 to 2:

$$I_\theta(r) = \int_0^2 (1-x)\sin(krx)\, dx$$

Integrating by parts yields:

$$I_\theta(r) = \left[-\frac{(1-x)\cos(krx)}{kr}\right]_0^2 - \frac{1}{kr}\int_0^2 \cos(krx)\, dx = \frac{1+\cos(2kr)}{kr} - \frac{\sin(2kr)}{(kr)^2}$$

Using the trigonometric identity $1+\cos(2kr) = 2\cos^2(kr)$, the force integral simplifies to:

$$F_{\rm drag} = -\frac{4\pi G^2 M^2 \rho_0}{v^2} \int_{r_{\rm min}}^{r_{\rm max}} \left[\frac{2\cos^2(kr)}{kr} - \frac{\sin(2kr)}{(kr)^2}\right] \frac{dr}{r}$$

The upper boundary $r_{\rm max} = b_{\rm max}$ corresponds to the finite physical spatial extent of the dark matter halo (or soliton core), while $r_{\rm min} = b_{\rm min}$ represents the effective inner cutoff scale. The integral over the oscillatory terms yields a quantum-modified Coulomb logarithm, $\ln\Lambda_Q$:

$$F_{\rm drag} = -\frac{4\pi G^2 M^2 \rho_0}{v^2} \ln\Lambda_Q$$

4.2 The High-Velocity Limit ($v \gg v_{\rm dB}$)

In the high-velocity regime, the perturber moves much faster than the characteristic de Broglie phase velocity of the background scalar field ($v \gg v_{\rm dB} \equiv \hbar/(m b_{\rm max})$). Consequently, the wavenumber $k = mv/\hbar$ becomes arbitrarily large, and the macroscopic de Broglie wavelength becomes negligible compared to the system size ($\lambda_{\rm dB} \ll b_{\rm max}$).

In this limit, the quantum wavenumber scale obeys $kr \gg 1$ over virtually the entire domain of integration. The rapidly oscillating terms $\cos(2kr)$ and $\sin(2kr)$ average to zero over macroscopic spatial scales:

$$\lim_{kr\to\infty} \left(\frac{2\cos^2(kr)}{kr} - \frac{\sin(2kr)}{(kr)^2} \right) \approx \frac{1}{kr}$$

Thus, the integral over radial impact parameters reduces to the classical logarithmic integral:

$$\int_{b_{\min}}^{b_{\max}} \frac{dr}{r} = \ln\left(\frac{b_{\max}}{b_{\min}}\right)$$

To account for both quantum and classical inner boundaries, the minimum impact parameter is defined as the quadrature sum of the classical impact parameter $b_{\rm cl} = GM/v^2$ (or physical perturber radius $r_{\rm p}$) and the effective quantum de Broglie scale $b_Q = \frac{\hbar}{2mv}$:

$$b_{\min} = \sqrt{r_{\rm p}^2 + \left(\frac{\hbar}{2mv}\right)^2}$$

When $v \gg v_{\rm dB}$, $b_Q \to 0$, and the drag force asymptotically matches Chandrasekhar's classical formula for a collisionless particle background:

$$\mathbf{F}_{\rm drag} \approx -\frac{4\pi G^2 M^2 \rho_0}{v^2} \ln\left(\frac{b_{\max}}{b_{\rm cl}}\right) \hat{\mathbf{v}}$$

This confirms that on large, high-velocity astrophysical scales (e.g., massive galaxy cluster mergers or high-velocity halo traversals), Fuzzy Dark Matter reproduces classical ΛCDM galactic dynamics.

4.3 The Low-Velocity Limit ($v \lesssim v_{\rm dB}$)

The fundamental divergence from classical dark matter physics occurs when the perturber slows to subsonic or sub-de Broglie velocities ($v \lesssim v_{\rm dB}$). As $v \to 0$, the de Broglie wavelength expands dramatically:

$$\lambda_{\rm dB} = \frac{2\pi\hbar}{mv} \gtrsim b_{\max}$$

When $\lambda_{\rm dB}$ expands to encompass the entire physical dimensions of the background soliton core ($b_Q \geq b_{\max}$), the lower bound of the Coulomb logarithm exceeds the upper integration limit ($b_{\min} \geq b_{\max}$).

Mathematically, for small arguments $kr \ll 1$, we expand $\sin(krx)$ to third order and integrate term by term:

$$\sin(krx) \approx krx - \frac{(krx)^3}{6}$$

$$I_\theta(r) = \int_0^2 (1-x)\left(krx - \frac{(krx)^3}{6}\right) dx = kr\int_0^2 (x-x^2)\,dx - \frac{(kr)^3}{6}\int_0^2 (x^3-x^4)\,dx$$

Evaluating each integral, $\int_0^2 (x-x^2)dx = -2/3$ and $\int_0^2 (x^3-x^4)dx = -12/5$, so

$$I_\theta(r) \approx -\frac{2}{3}kr + \frac{2}{5}(kr)^3$$

This matches the small-kr expansion of the exact closed form derived in Section 4.1, $I_\theta(r) = \frac{1+\cos(2kr)}{kr} - \frac{\sin(2kr)}{(kr)^2}$, confirming the coefficient.

Substituting this back into the radial force integral yields:

$$F_{\rm drag} = -\frac{4\pi G^2 M^2 \rho_0}{v^2}\int_0^{b_{\max}} \frac{I_\theta(r)}{r}\,dr \approx -\frac{4\pi G^2 M^2 \rho_0}{v^2}\int_0^{b_{\max}} \left(-\frac{2}{3}k + \frac{2}{5}k^3 r^2\right) dr$$

To leading order in k, this is

$$F_{\rm drag} \approx \frac{4\pi G^2 M^2 \rho_0}{v^2}\cdot\frac{2}{3}k\,b_{\max} \;\propto\; k \;\propto\; v$$

The overall negative sign of the leading term in $I_\theta(r)$ combines with the pre-existing minus sign in the force integral to give a positive, physically decelerating drag force, consistent with dynamical friction acting to oppose the perturber's motion. Instead of diverging as $1/v^2$ as in classical Chandrasekhar theory, the drag force scales linearly with velocity as $v \to 0$, forcing $F_{\rm drag} \to 0$.

5. Orbital Dynamics in Solitonic Cores

5.1 Hamiltonian Formulation of the Orbit

To model the full trajectory of a massive perturber (such as a globular cluster or dwarf satellite) moving through the Fuzzy Dark Matter (FDM) halo, we formulate its mechanics within a non-conservative Hamiltonian framework. The perturber of mass M moves through the background gravitational potential of the solitonic core, $\Phi_c(\mathbf{r})$, while simultaneously

experiencing the dissipative drag force $\mathbf{F}_{\text{drag}}(\mathbf{v}, \mathbf{r})$ derived from the quantum wave wake.

The Hamiltonian for the conservative portion of the system is standard:

$$H(\mathbf{r}, \mathbf{p}) = \frac{\mathbf{p}^2}{2M} + M\Phi_c(\mathbf{r})$$

Deep within the solitonic core ($r \ll r_c$), the FDM density is approximately constant, $\rho(\mathbf{r}) \approx \rho_0$. Integrating the Poisson equation $\nabla^2\Phi_c = 4\pi G\rho_0$ yields a spherically symmetric, harmonic oscillator-like potential:

$$\Phi_c(r) \approx \frac{2\pi}{3} G\rho_0 r^2$$

Because the system is open and kinetic energy is continually radiated away into the FDM field as scalar waves, the Hamiltonian is not conserved. The phase-space evolution is governed by the modified Hamilton's equations, where the generalized momentum incorporates the non-conservative drag:

$$\dot{\mathbf{r}} = \frac{\mathbf{p}}{M}$$

$$\dot{\mathbf{p}} = -\nabla_{\mathbf{r}} H + \mathbf{F}_{\text{drag}}\left(\frac{\mathbf{p}}{M}, \mathbf{r}\right) = -M\omega_0^2\mathbf{r} + \mathbf{F}_{\text{drag}}$$

where $\omega_0 = \sqrt{\frac{4\pi}{3} G\rho_0}$ is the natural orbital frequency of the soliton core. The instantaneous rate of energy dissipation is given by the power drawn by the quantum wake:

$$\frac{dH}{dt} = \dot{\mathbf{r}} \cdot \dot{\mathbf{p}} + \frac{\partial H}{\partial t} = \mathbf{v} \cdot \mathbf{F}_{\text{drag}}$$

5.2 Numerical Integration of Orbital Decay

Analytic solutions to these equations of motion are unavailable due to the highly nonlinear velocity dependence of $\mathbf{F}_{\text{drag}}$ across the transition from the classical to the quantum regime (from $1/v^2$ to linear v). Therefore, the trajectories must be integrated numerically.

A semi-implicit symplectic integration scheme (e.g., a modified leapfrog algorithm) is typically employed to preserve the geometric properties of the conservative harmonic motion while accurately capturing the gradual phase-space contraction caused by the dissipative drag term.

1. **Infall Phase:** When the perturber enters the halo from a large radius ($r \gg r_c$), its orbital velocity is much greater than the local de Broglie velocity ($v \gg v_{\rm dB}$). In this phase, $\mathbf{F}_{\rm drag} \propto 1/v^2$. The perturber rapidly dissipates angular momentum and spirals inward. Phase-space volume contracts rapidly, violating Liouville's theorem as expected for a dissipative system.

2. **Transition Phase:** As the perturber crosses into the soliton core, its potential energy drops, but because the enclosed mass $M_{\rm enc}(r) \propto r^3$ declines faster than r, its orbital velocity actually decreases.

3. **Stalling Phase:** Once the orbital velocity drops to $v \sim v_{\rm dB}$, the spatial extent of the wave wake $\lambda_{\rm dB}$ expands to encompass the perturber. The drag force transitions to the quantum-suppressed regime ($\mathbf{F}_{\rm drag} \propto v$). The rate of energy dissipation exponentially decays, and the orbit stabilizes.

Unlike classical ΛCDM numerical simulations—where the perturber inevitably spirals into a singular density cusp at $r = 0$ (the "core collapse")—the FDM numerical integration reveals that the phase-space attractor is a finite, stable limit cycle.

5.3 Derivation of the Stalling Radius ($r_{\rm stall}$)

We can derive a rigorous analytical estimate for the radius at which this orbital decay definitively halts, known as the stalling radius $r_{\rm stall}$.

For a perturber on a circular orbit within the near-constant density region of the soliton core, the centripetal force is provided entirely by the background harmonic potential. The classical orbital velocity is:

$$v_{\rm orb}(r) = \sqrt{r\frac{d\Phi_c}{dr}} = \sqrt{\frac{4\pi}{3}G\rho_0 r^2} = \omega_0 r$$

As established in Section 4.3, dynamical friction is fundamentally suppressed when the perturber's velocity falls below the effective phase velocity of the background scalar field, $v_{\rm dB} \equiv \hbar/(mr)$. Stalling occurs at the critical radius where the local orbital velocity equals this quantum velocity threshold:

$$v_{\rm orb}(r_{\rm stall}) \approx \frac{\hbar}{m r_{\rm stall}}$$

Substituting the expression for the orbital velocity:

$$\sqrt{\frac{4\pi}{3}G\rho_0}\, r_{\rm stall} \approx \frac{\hbar}{m r_{\rm stall}}$$

Solving for $r_{\rm stall}^2$ yields:

$$r_{\rm stall}^2 \approx \frac{\hbar}{m}\sqrt{\frac{3}{4\pi G\rho_0}}$$

$$r_{\rm stall} \approx \left(\frac{3\hbar^2}{4\pi G\rho_0 m^2}\right)^{1/4}$$

This result represents a profound physical consistency within the FDM paradigm. By comparing this derived kinematic stalling radius to the static structural properties of the scalar field derived in Section 2, we observe that $r_{\rm stall}$ is strictly proportional to the Quantum Jeans length:

$$\lambda_J = \left(\frac{\pi^3\hbar^2}{G\rho_0 m^2}\right)^{1/4}$$

Therefore, $r_{\rm stall} \sim \lambda_J \sim r_c$. The perturber ceases its inward spiral precisely at the boundary of the soliton core.

6. Observational Constraints: The Fornax Dwarf Spheroidal

6.1 Globular Cluster Kinematics in Fornax

The theoretical mechanisms of quantum-suppressed dynamical friction find their most rigorous astrophysical test bed in the Fornax dwarf spheroidal (dSph) galaxy. Fornax is a dark-matter-dominated satellite of the Milky Way, distinguished by its population of five ancient, massive globular clusters (GCs). Observations reveal that these clusters reside at projected distances ranging from 0.24 kpc to 1.6 kpc from the center of the galaxy.

In the standard Cold Dark Matter (ΛCDM) paradigm, the dark matter halo follows a cuspy Navarro-Frenk-White (NFW) profile. A massive body moving through this dense, classical background experiences the unsuppressed $1/v^2$ Chandrasekhar drag force. Given the masses of the Fornax GCs ($M \sim 10^5 M_\odot$) and the central density of the Fornax halo, the classical

dynamical friction timescale $t_{\rm df}$ for these clusters to lose their orbital angular momentum and sink to the galactic center is on the order of 2 to 3 Gyr.]

Because the stellar populations of these globular clusters indicate ages exceeding 10 Gyr, they should have spiraled into the center of Fornax billions of years ago, forming a dense nuclear star cluster. Observations conclusively show that Fornax possesses no such nuclear cluster. This severe discrepancy between theoretical prediction and observational reality is known as the Fornax Timing Problem, and it stands as one of the most stubborn failures of classical ΛCDM on sub-galactic scales.

6.2 Applying the FDM Stalling Model

The Fuzzy Dark Matter (FDM) framework completely resolves the Fornax Timing Problem not by fine-tuning the cluster initial conditions, but by fundamentally altering the nature of the dark matter background via the quantum pressure tensor.

As derived in Section 5, as a globular cluster decays and approaches the inner region of the halo, it enters the solitonic core. Here, the local velocity of the cluster drops as it orbits within the flat-density harmonic potential ($\Phi_c \propto r^2$). Simultaneously, the effective de Broglie wavelength of the background field expands.

When the cluster's orbital velocity satisfies $v_{\rm orb} \lesssim v_{\rm dB}$, the quantum wave wake is exponentially suppressed. The dense, trailing classical wake that normally provides the backward gravitational pull is smoothed out by the FDM quantum pressure, leaving a symmetric density cavity around the perturber. Consequently, the drag force undergoes a dramatic transition from $F_{\rm drag} \propto 1/v^2$ to $F_{\rm drag} \propto v$, and the energy dissipation rate plummets.

The globular clusters asymptotically approach the stalling radius $r_{\rm stall}$. The FDM model predicts that ancient massive satellites will be found "parked" exactly at the boundary of the solitonic core. The observed spatial distribution of the Fornax GCs at ~ 1 kpc therefore corresponds to a permanent, dynamically stable equilibrium.

6.3 Constraining the Boson Mass

Because the stalling radius is mathematically inextricably linked to the fundamental parameters of the scalar field, the observed positions of the Fornax globular clusters provide a powerful astrophysical laboratory for constraining the FDM boson mass m.

From Section 5.3, the analytic stalling radius is strictly tied to the central density ρ_0 and the mass of the scalar particle:

$$r_{\rm stall} \approx \left(\frac{3\hbar^2}{4\pi G \rho_0 m^2} \right)^{1/4}$$

By rearranging this relation, we can isolate the fundamental particle mass m as a function of macroscopic galactic observables:

$$m \approx \frac{\hbar}{r_{\rm stall}^2} \sqrt{\frac{3}{4\pi G \rho_0}}$$

To apply this to Fornax, we utilize standard kinematic data. The innermost globular clusters are stalled at roughly $r_{\rm stall} \sim 1$ kpc. The central core density of Fornax, derived from stellar velocity dispersion profiles, is approximately $\rho_0 \sim 0.1\, M_\odot\, {\rm pc}^{-3}$.

Inserting these astrophysical values into the stalling equation yields a predicted boson mass of:

$$m \sim 10^{-22}\ {\rm eV}$$

This result is profoundly significant. A scalar field with $m \sim 10^{-22}$ eV is exactly the mass range required to generate a kiloparsec-scale solitonic core and suppress the matter power spectrum at small scales, resolving the Core-Cusp and Missing Satellites problems. The fact that globular cluster stalling kinematics in Fornax independently select the exact same ultralight mass scale represents a highly non-trivial cross-validation of the FDM paradigm.

7. CONCLUSION

7.1 Summary of Analytical Findings

In this work, we have systematically investigated the orbital dynamics of massive perturbers within the Fuzzy Dark Matter (FDM) paradigm,

bridging microscopic quantum wave mechanics with macroscopic galactic kinematics. By modeling the dark matter halo as a self-gravitating Bose-Einstein condensate governed by the coupled Schrödinger-Poisson equations, we demonstrated that the classical Chandrasekhar theory of dynamical friction breaks down at the boundary of the central solitonic core. Using linear perturbation theory under the Born approximation, we derived the explicit structural form of the FDM gravitational wake $\delta\rho(\mathbf{r})$ generated by a moving mass M. Unlike the singular $1/r$ wake of classical collisionless Cold Dark Matter (ΛCDM), the FDM wake manifests as a continuous, parabolic wave interference pattern where density fluctuations are smoothed out within a de Broglie wavelength $\lambda_{\rm dB} = 2\pi\hbar/mv$.

7.2 Cosmological Implications

These findings offer a candidate resolution to several small-scale crises that plague standard ΛCDM astrophysics. In particular, the wave-mechanical stalling of dynamical friction offers a mechanism to address the long-standing Fornax Globular Cluster Timing Problem without requiring fine-tuned, stochastic baryonic feedback to erase central dark matter cusps or alter cluster orbits. This resolution is not unconditional: full numerical treatments of FDM dynamical friction (Lancaster et al., 2020) find that the stalling mechanism ceases to rescue the Fornax timing problem for boson masses $m \gtrsim 10^{-21}$ eV, so the viability of this explanation is contingent on the boson mass lying within the range independently favored by the soliton-core scale, rather than holding as a parameter-free result. Within that range, FDM offers a natural account of the survival and current spatial arrangement of Fornax's ancient globular clusters at ~ 1 kpc as a stable phase-space equilibrium.

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